EMCC 2026

Solutions



January 18, 2026

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Organizing Acknowledgments

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Chapter 1

Speed Test Solutions



1. What is the average of 9999 and 100001?

Solution: The answer is $\boxed{55000}$.
The average of 9999 and 100001 equals $\frac{9999+100001}{2} = \frac{110000}{2} = 55000$.

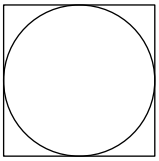
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2. What is $\sqrt{36 \times 49}$?

Solution: The answer is $\boxed{42}$.
We have $\sqrt{36 \cdot 49} = \sqrt{36} \cdot \sqrt{49} = 6 \cdot 7 = 42$.

---

3. A circle is inscribed in a square, as shown below. If the area of the square and the circumference of the circle are both equal to x, what is x?



Solution: The answer is $\boxed{\pi^2}$.
Let the side length of the square be s. The diameter of the circle is also s, so we have that $x = s^2 = s\pi$. Dividing by s gives that $s = \pi$, so $x = s^2 = \pi^2$.

---

4. Grant studies for 6 hours and 26 minutes, beginning a minutes after midnight and finishing b minutes after 7:00 AM. What is $a - b$?

Solution: The answer is $\boxed{34}$.
We have that

$$a + (6 \cdot 60 + 26) - b = 7 \cdot 60,$$

so $a - b = 60 - 26 = 34$.

---

5. If

$$x + (x+1) + (x+2) + (x+3) + (x+4) = 100 + 102 + 104 + 106 + 108,$$

what is x?

Solution: The answer is $\boxed{102}$.

The LHS and RHS of the equation are each the sums of five terms in an arithmetic progression. Therefore, their middle terms should be equal. So, we have $x + 2 = 104$, and $x = 102$.

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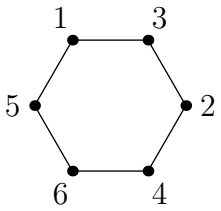
6. What is the largest integer dividing 432, 324 and 243?

Solution: The answer is $\boxed{27}$.

By 2026 OTIS Mock AIME #1, we have that $\gcd(432, 324) = 108$. Clearly, $\gcd(108, 243) = 27$.

---

7. The numbers 1 through 6 are written at the vertices of a hexagon below. In a move, Alice can swap any two numbers. What is the fewest number of swaps necessary for the numbers to be sorted in either clockwise or counterclockwise order?



Solution: The answer is $\boxed{1}$.

Swapping the 1 and 4 sorts the numbers in counterclockwise order, so the answer is 1.

---

8. What fraction of the numbers in $\{1, 2, 3, \ldots, 899\}$ are perfect squares?

Solution: The answer is $\boxed{\frac{1}{31}}$.

Since $30^2 = 900$, there are 29 perfect squares in $\{1, 2, 3, \ldots, 899\}$. We can use difference of squares to write $899 = 30^2 - 1 = (30 + 1)(30 - 1) = 31(29)$, so $\frac{29}{899} = \frac{29}{29(31)} = \frac{1}{31}$.

---

9. Alice and Bob are attendees at a party with 10 people total. Each person randomly chooses somebody else at the party to gift a present. What is the probability that somebody gets a present from both Alice and Bob?

Solution: The answer is $\boxed{\frac{8}{81}}$.

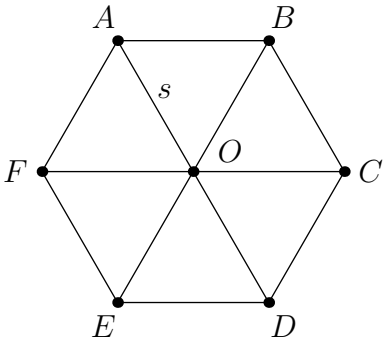
Since Alice and Bob cannot gift to themselves, this person must be one of the other 8 attendees. The probability that Alice gives a gift to one of these attendees is $\frac{8}{9}$. Then Bob must choose the same person, which occurs with probability $\frac{1}{9}$, so the answer is $\frac{8}{9} \cdot \frac{1}{9} = \frac{8}{81}$.

---

10. An equilateral triangle with area 10 has side length s. What is the area of a regular hexagon with circumradius s?

Solution: The answer is $\boxed{60}$.

We can put six of these equilateral triangles together as such:



Note that $ABCDEF$ is the hexagon we are looking for, as point O is a distance of s from each vertex. Its area is six times that of the triangle, giving us 60.

---

11. Jesse is purchasing gas masks. He doesn't know the price of one gas mask, but he knows it costs \$274.32 to buy 9 gas masks. What is the fewest positive number of gas masks that Jesse must buy so that the total is an integer number of dollars?

Solution: The answer is $\boxed{25}$.

By looking at the last two digits, we see that 27432 is a multiple of 4 but not 5. Dividing by 9 does not change this fact. Therefore, the smallest integer which we can multiply $\frac{27432}{9}$ by to get an integer is 25.

---

12. Let ℓ be a line with slope 2 and let m be a line with slope -1. Lines ℓ, m and x-axis determine a triangle with an area of 108. How far apart are the x-intercepts of ℓ and m?

Solution: The answer is $\boxed{18}$.

Suppose the two lines meet at a point with y-coordinate h. We can also shift the lines left or right so their intersection has x-coordinate 0. Then, since ℓ has slope 2, it meets the x-axis at $(-\frac{h}{2}, 0)$. Since m has slope -1, it meets the x-axis at $(h, 0)$. So, the area of the triangle is

$$\left(\frac{3h}{2}\right) h.$$

Setting this equal to 108, we find that $h = 12$, so $\frac{3}{2}h = 18$, as desired.

---

13. When an unfair coin is flipped 10 times, there is a $\frac{2}{3}$ chance that the total number of heads is even. If, instead, the coin was flipped 20 times, what is the probability that the total number of heads is even?

Solution: The answer is $\boxed{\frac{5}{9}}$.

Split the 20 flips into two groups of 10. We require that there is an even number of heads in both groups or an odd number of heads in both groups. This occurs with probability $\left(\frac{2}{3}\right)^2 + \left(\frac{1}{3}\right)^2 = \frac{5}{9}$.

---

14. What is the sum of the digits in the decimal expansion of 5^{-8}?

Solution: The answer is $\boxed{13}$.

We have that $5^{-8} = \frac{2^8}{10^8} = 256 \cdot 10^{-8}$, so the sum of digits will be $2 + 5 + 6 = 13$.

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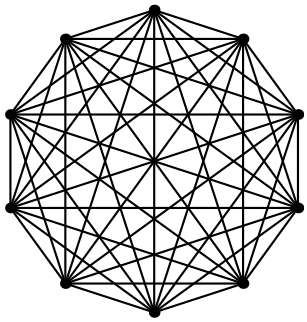
15. In isosceles triangle ABC with $AB = AC$, the incircle passes through the centroid. What is AB/BC? (The *centroid* of a triangle is the common intersection point of its three medians.)

Solution: The answer is $\boxed{\frac{5}{2}}$.

Let I be the incenter, let D be the foot from A to' $\overline{BC}$ and let E be the foot from I to $\overline{AC}$. Since $AG : GD = 2 : 1$ and I is the midpoint of $\overline{GD}$, it follows that $AI : ID = 5 : 1$. So, $AI : IE = 5 : 1$. By AA similarity, we have $\triangle AIE \sim \triangle ABD$, so $\frac{AB}{BC} = \frac{AB}{2BD} = \frac{5}{2}$.

---

16. Lotad draws the sides and diagonals of a regular 10-sided polygon. Then, she draws a line ℓ that does not pass through any vertices in the polygon. What is the maximum possible number of sides and diagonals that ℓ could intersect?



Solution: The answer is $\boxed{25}$.

The line l will partition the vertices of the polygon into two sets, one on each side of the line. A given side or diagonal will intersect l if and only if its endpoints lie on opposite sides of the line. The number of intersections is thus the product of the sizes of the two sets. Since there are 10 total elements, by AM-GM this product is maximized when there are 5 on each side, which gives a total of $5^2 = 25$ intersections.

---

17. A *repdigit* is a positive integer whose digits are all the same. What is the maximum number of distinct digits that the absolute difference of two repdigits could have?

Solution: The answer is $\boxed{4}$.

The difference of 1111 and 22 is 1089, with four distinct digits. It remains to show that this is the best bound possible.

Suppose the larger repdigit consists of repeated A's and the smaller repdigit consists of repeated B's. If $A \geq B$, every digit in the difference must be either $A - B$ or A, so there are at most 2 distinct digits. If $A < B$, every digit in the difference must be either $A - B + 10$, $A - B + 9$, A or $A - 1$, so there are at most 4 distinct digits.

---

18. A 7-digit integer has strictly increasing digits from left to right. How many such integers are divisible by 3?

Solution: The answer is $\boxed{12}$.

Zero cannot be the first digit and thus cannot appear in the number due to the strictly increasing condition. We are wish to choose 7 numbers from 1 through 9 which sum to a multiple of 3, or equivalently, choose two numbers summing to a multiple of 3 to leave out, because $1 + 2 + \cdots + 9 = 45$

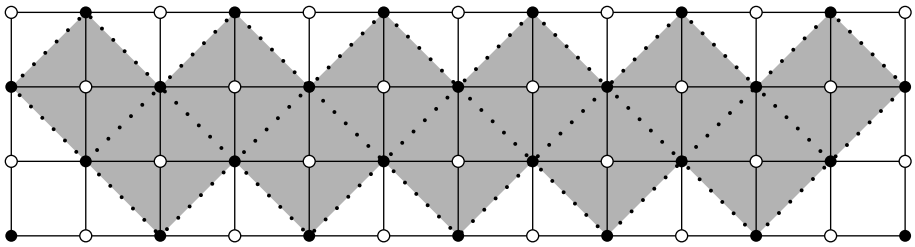
is divisible by 3. This can occur is one numbers is 1 (mod 3) and the other is 2 (mod 3), or if they are both divisible by 3, for a total of $3(3) + \binom{3}{2} = 9 + 3 = 12$ cases.

---

19. In a 3 by 12 grid of unit squares, Albert wishes to draw diagonals in some of the unit squares such that the diagonals form a single cycle that does not cross or touch itself. How many ways can Albert draw these diagonals?

Solution: The answer is $\boxed{132}$.

Albert's path encloses a polyomino made from $\sqrt{2} \times \sqrt{2}$ squares. Moreover, if we color every vertex of the grid black or white in checkerboard fashion, each $\sqrt{2} \times \sqrt{2}$ square has vertices of the same color.



We will count the number of paths on black vertices and multiply by 2 at the end. The set of $\sqrt{2} \times \sqrt{2}$ squares with black vertices forms a tilted staircase, as shown above. To form a polyomino within this 11-cell staircase, it suffices to choose the leftmost cell and the rightmost cell to include in the polyomino. By doing casework on whether these cells are distinct or not, there are $\binom{11}{2} + 11 = 66$ polyominos.

Remembering to multiply by 2 in the case that the vertices are white instead of black, we obtain our final answer of 132.

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20. Let $PA_1A_2 \cdots A_{10}$ be a pyramid such that base $A_1A_2 \cdots A_{10}$ is a regular 10-sided polygon. There exists a sphere tangent to every face of the pyramid, as well as a sphere passing through every vertex of the pyramid. Given that these two spheres share a center, what is $\angle PA_1A_2$, in degrees?

Solution: The answer is $\boxed{81^\circ}$.

We claim that every face has the same circumradius. This would imply that $\angle A_1PA_2 = 18^\circ$ by the inscribed angle theorem, so $\angle PA_1A_2 = 81^\circ$. (Evidently, the triangular faces are all isosceles.)

To show the claim, let O be the coinciding point of the incenter and circumcenter. Let R and r be the circumradius and inradius of the pyramid. For each face $\mathcal{F}$ of the pyramid, let O' be the foot from O to $\mathcal{F}$. So, if X is any vertex of $\mathcal{F}$, we have that $\triangle O'OX$ is a right triangle. We know that $OO' = r$ and that $OX = R$, so by the Pythagorean theorem, $O'X = \sqrt{R^2 - r^2}$. Repeating this argument on all vertices X of $\mathcal{F}$, we see that $\mathcal{F}$ is cyclic with circumradius $\sqrt{R^2 - r^2}$ and center O'. Repeating this argument on all faces $\mathcal{F}$ implies the claim.


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Chapter 2

Accuracy Test Solutions



1. At a clothing store, sweaters usually cost \$56.22 more than jeans. With a 40% discount, sweaters now cost \$56.22. How many jeans are equal to the price of 30 discounted sweaters?

Solution: The answer is $\boxed{45}$.

Let $x = \$56.22$. The second sentence tells us that x is equal to 60% of the price of a non-discounted sweater y, so the first sentence tells us that jeans cost $0.4y$. Then we get that the price of 30 discounted sweaters is $30 \cdot 0.6y = 45 \cdot 0.4y$, so the answer is 45.

---

2. Sabrina rolls a fair six-sided die three times. If the product of the the three numbers obtained has fifteen divisors, what is the sum of the three numbers?

Solution: The answer is $\boxed{16}$.

The number must be of the form p^{14} or p^4q^2 for distinct primes p and q. The first option is clearly impossible with only three rolls. The second option is only achievable with rolls of 4, 6, and 6 in some order, leading to a sum of 16.

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3. A right triangle $\mathcal{T}$ has side lengths a, b and c. If $abc = 260$ and $a^2 + b^2 + c^2 = 200$, what is the area of $\mathcal{T}$?

Solution: The answer is $\boxed{13}$.

Let the hypotenuse be c. Then $(a^2 + b^2) + c^2 = 2c^2 = 200$ implies that $c = 10$. Then the area of the triangle is $\frac{ab}{2} = \frac{260}{2c} = 13$.

---

4. Suppose A, B, C and D are (not necessarily distinct) digits such that the two-digit numbers $\underline{AB}$, $\underline{BC}$ and $\underline{CD}$ are consecutive (in that order). What is the four digit number $\underline{ABCD}$?

Solution: The answer is $\boxed{8890}$.

Since $\underline{AB}$ and $\underline{BC}$ are consecutive, their units digits B and C cannot be equal. This means that the tens digits of $\underline{BC}$ and $\underline{CD}$ are not equal, so we must have $C = 9$ and $D = 0$. From there we fill in $A = B = 8$ to get 8890, which works.

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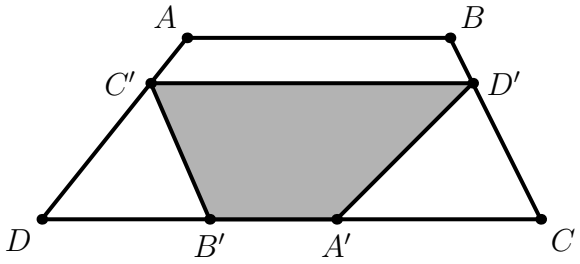
5. Michael has a 9×9 square of chocolate. Every minute, he takes a rectangle of chocolate and splits it along a vertical or horizontal line into two pieces of chocolate, each with integer side lengths. When Michael is unable to split any more pieces of chocolate, he stops. What is the average length of the cuts that Michael makes?

Solution: The answer is $\boxed{\frac{9}{5}}$.

Each cut increases the total number of pieces by 1. Since Michael stops with 81 unit squares of chocolate, he makes 80 cuts. The total length of cuts that he makes is the sum of the length of the 8 vertical and 8 horizontal line segments in the interior of the grid. Thus the average cut length is $\frac{(8+8)(9)}{80} = \frac{9}{5}$.

---

6. Let $ABCD$ be a trapezoid with $\overline{AB} \parallel \overline{CD}$. A trapezoid $A'B'C'D'$ similar to $ABCD$ is inscribed inside, as shown below. Given that $AB = 6$, $BD' = 1$ and $D'C = 3$, what is CA'?



Solution: The answer is $\boxed{\frac{9}{4}}$.

Since $\frac{CD'}{CB} = \frac{3}{4}$, the ratio of the heights of the trapezoids is also $\frac{3}{4}$. By the similarity, we get that $B'A' = 6 \cdot \frac{3}{4} = \frac{9}{2}$. The similarity also tells us that $AD \parallel D'A'$ and $BC \parallel C'B'$, and the parallelograms this implies that $C'D' = DA' = B'C$.

To finish, we let $x = CA'$ and note that $C'D'$ and DC are in a $3 : 4$ ratio as well. We obtain the equation

$$\frac{x + \frac{9}{2}}{2x + \frac{9}{2}} = \frac{3}{4}.$$

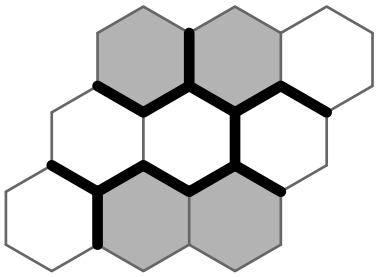
Expanding and solving gives $x = \frac{9}{4}$.

---

7. In the hexagonal grid below, some edges are bolded, dividing the grid into six regions. Each of the nine cells is to be filled with a distinct single digit integer, such that

   - no consecutive numbers share a **bolded** edge;
   - the one or two digit number formed by each region is a prime number.

   What is the product of the digits in the four shaded cells?



Solution: The answer is $\boxed{672}$.

Label the first row ABC, the second row DEF and the last row GHI.

(a) The digits 4, 6 and 8 can only appear in spots B, D or H since they cannot be the units digit of a prime. In particular, BDH must be some permutation of 468.

(b) The digit 5 cannot appear in CEI, so it must be one of the single digit primes. Both A and G will border two numbers from $\{4, 6, 8\}$, so neither of them can be 5. Therefore, F is 5.

(c) Since B cannot be 4 or 6, it must be 8.

(d) Note that DH is a permutation of 46. So, G cannot be 3 or 7, so it must be 2.

(e) Since A touches 8, it must be 3.

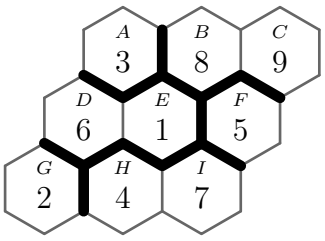
(f) Since D touches 3, it must be 6. Now, H must be 4.

(g) Since E touches 8, it must be 1.

(h) Since the number $\overline{4I}$ is prime, I must be 7.

(i) This leaves us with C, which is 9. (Alternatively, after deduction (a), since 9, 49 and 69 are composite, we already know that $C = 9$.)

We extract $3 \times 8 \times 4 \times 7 = 672$.



8. Let $A_1A_2 \ldots A_9$ be a regular 9-sided polygon. Two points X and Y are randomly selected inside the polygon. What is the probability that segments A_1X and A_2Y intersect?

Solution: The answer is $\boxed{\frac{7}{18}}$.

Suppose that $A_1A_2 \cdots A_9$ has area 1. Then, by symmetry, the expected value of $[A_1A_2X]$ is $\frac{1}{9}$. So, the probability that Y lies inside of $\triangle A_1A_2X$ is $\frac{1}{9}$. Likewise, the probability that X lie inside of $\triangle A_1A_2Y$ is $\frac{1}{9}$.

These events are mutually exclusive, and in the $\frac{7}{9}$ chance that neither is true, then either A_1A_2XY or A_1A_2YX is a convex quadrilateral.

Again, these two events are mutually exclusive, so they must both occur with a $\frac{7}{18}$ probability. Lines A_1X and A_2Y intersect if and only if A_1A_2XY is convex, which we know is a $\frac{7}{18}$ chance.

---

9. If a, b, c and d are positive integers such that $\gcd(a! + b!, c! + d!) = 1320$, what is $a \times b \times c \times d$?

Solution: The answer is $\boxed{4050}$.

Claim: We have $\min(a, b, c, d) = 5$.

Proof: Let m stand as shorthand for $\min(a, b, c, d)$. Since $m!$ divides each of $a!$, $b!$, $c!$ and $d!$, we have $m \leq 5$. Moreover, $5!$ dividing $a! + b!$ implies that $a, b \geq 5$. Likewise, $c, d \geq 5$, so $m = 5$.

Without loss of generality, set a to 5. We can easily compute that the remainders when $5!, \ldots, 10!$ are divided by 11 are 10, 5, 2, 5, 1 and 10, respectively. Since we have $11 \mid a! + b!$, this implies that $b = 9$. We have that

$$5! + 9! = 120(1 + 6 \cdot 7 \cdot 8 \cdot 9) = 120(1 + 54 \cdot 56) = 120 \cdot 55^2.$$

Since $5^2 \nmid 1320$, it follows that $5^2 \nmid c! + d!$. So, $\min(c, d) \leq 9$. This also implies that $\max(c, d) \leq 10$ (because of divisibility by 11). From the fact that $11 \mid c! + d!$, we now see that $\{c, d\} = \{9, 10\}$. Our final answer is $5 \times 9 \times 9 \times 10 = 4050$.

---

10. Let a, b and c be positive real numbers and let $P(x) = x^3 - ax^2 + bx - c$ be a polynomial with real roots $\frac{1}{3}$, r and s. If $c^2 = aP(\frac{a}{2})$, what is the maximum possible value of $r^2 - s^2$?

Solution: The answer is $\boxed{\frac{\sqrt{2}}{6}}$.

By Descartes' rule of signs, r and s are positive. Note that

$$K = \sqrt{\frac{a}{2} P\left(\frac{a}{2}\right)}$$

is the area of a triangle with side lengths $\frac{1}{3}$, r and s. Let the circumradius of this triangle be R. By geometric formulas, we have that $\frac{rs}{3} = 4KR$. So, we have

$$\begin{aligned} c &= \sqrt{2}\sqrt{\frac{a}{2}P\left(\frac{a}{2}\right)} \\ \frac{rs}{3} &= K\sqrt{2} \\ 4KR &= K\sqrt{2} \\ R &= \frac{\sqrt{2}}{4}. \end{aligned}$$

So, our problem reduces to this: we have a $\triangle ABC$ with circumradius $\frac{\sqrt{2}}{4}$; we are given that $BC = \frac{1}{3}$, and we wish to maximize $AB^2 - AC^2$. Let D be the foot from A to $\overline{BC}$, and let M be the midpoint of $\overline{BC}$. We have

$$\begin{aligned} AB^2 - AC^2 &= (AD^2 + DB^2) - (AD^2 + DC^2) \\ &= DB^2 - DC^2 \\ &= 2(BC)(DM). \end{aligned}$$

Let O be the circumcenter of $\triangle ABC$. Since M and D are the feet from O and A to $\overline{BC}$, respectively, we have $DM \leq OA = \frac{\sqrt{2}}{4}$, so

$$AB^2 - AC^2 \leq 2(BC)(DM) \leq \frac{\sqrt{2}}{6}.$$

Equality holds when $\overline{OA} \parallel \overline{BC}$.

Chapter 3

Team Test Solutions



1. Let N be a two-digit integer. If the sum of the digits of N is a multiple of 6 and the product of the digits of N is a multiple of 7, what is the sum of all possible values of N?

Solution: The answer is $\boxed{192}$.

For the product of digits to be a multiple of 7, at least one of the digits must be 0 or 7. Combined with the first condition, we obtain the three possibilities 60, 57, and 75 which sum to 192.

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2. When

$$(0.565656\cdots) + (0.134134134\cdots)$$

is written as a decimal, what is the sum of the distinct digits that appear after the decimal point?

Solution: The answer is $\boxed{22}$.

We can see that the sum will repeat every six digits. This repeating string will be $565656 + 134134 = 699790$, so the sum of the distinct digits is $6 + 9 + 7 + 0 = 22$.

---

3. In square $ABCD$ with side length 1, points X and Y lie on the incircle. If $AXCY$ is a rhombus, what is its area?

Solution: The answer is $\boxed{\frac{\sqrt{2}}{2}}$.

Because $AX = XC$ and $AY = YC$, both X and Y must lie on the perpendicular bisector of AC, which is BD. Then since BD passes through the center of the square, XY is a diameter of the incircle, so $XY = 1$. Then since the diagonals of a rhombus are perpendicular, its area is $\frac{1}{2} \cdot AC \cdot XY = \frac{\sqrt{2}}{2}$.

---

4. Grant thinks of a set of integers. He writes down all sums of distinct pairs of integers from the set. The smallest number he writes is -5 and the largest number he writes is 12. What is the maximum possible sum of the numbers in Grant's set?

Solution: The answer is $\boxed{17}$.

The second largest number in Grant's set is at most 5, or else he would write a number greater than 12. Then the third number is at most 4, so the greatest possible sum of the positive numbers in his set is $12 + 4 + 3 + 2 + 1 = 22$. Then the negative numbers in his set must sum to -5 or less in order for him to write -5, so the sum can be no more than $22 - 5 = 17$. The set $\{-3, -2, 1, 2, 3, 4, 5, 7\}$ achieves this.

---

5. What positive integer n satisfies $n + \text{lcm}(n, 30) = 576$?

Solution: The answer is $\boxed{96}$.

Since $n = 576 - 30k$ for some integer k, n is a multiple of 6 but not a multiple of 5. This means that $\gcd(n, 30) = 6$, so $\text{lcm}(n, 30) = \frac{30n}{6} = 5n$. Thus $6x = 576$ and $x = 96$.

---

6. What is the smallest positive real x such that $\lfloor x^2 \rfloor + \lfloor x \rfloor$ is a positive multiple of 5?

Solution: The answer is $\boxed{2\sqrt{2}}$.

When $x < 2$, $\lfloor x^2 \rfloor + \lfloor x \rfloor \leq 3 + 1 = 4$, so we must have $x \geq 2$. Then if $\lfloor x \rfloor = 2$, then $\lfloor x^2 \rfloor = 8$ is the only possible way to achieve a multiple of 5. This gives a minimum value of $2\sqrt{2}$.

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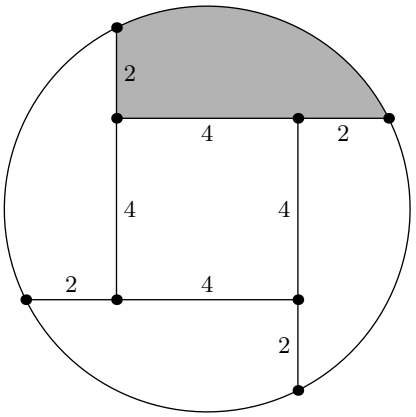
7. Grant thinks of some percentages summing to 100. He rounds each percentage to the nearest integer and writes the results down. If the new percentages sum to 97, what is the fewest number of percentages Grant could have written down?

Solution: The answer is $\boxed{7}$.

Rounding a percentage decreases it by strictly less than 0.5, so to decrease the total by 3, Grant must have rounded down at least 7 percentages.

---

8. Four segments are drawn inside a circle, forming right angles with each other. Given the side lengths in the diagram below, what is the area of the shaded region?



Solution: The answer is $\boxed{5\pi - 4}$.

By symmetry, the two points where two opposing segments meet the circle are diametrically opposite each other. This means that the diameter of the circle is $\sqrt{(2+4+2)^2+4^2} = 4\sqrt{5}$ and the area of the circle is $(2\sqrt{5})^2\pi = 20\pi$. Then if the area of the shaded region is x, we have that $4x + 16 = 20\pi$. Solving, we get $x = 5\pi - 4$.

---

9. Let $\mathcal{S}$ be the set of all positive integers n for which each prime divisor of n is less than 2026, and no perfect square greater than 1 divides n. Over all elements of $\mathcal{S}$, what is the average units digit?

Solution: The answer is $\boxed{\frac{15}{4}}$.

The units digit is 0 when both 2 and 5 divide n, which occurs $\frac{1}{4}$ of the time. For the rest of the cases, we will show that the average digit is exactly 5. For each term n not divisible by 19, we pair n with $19n$. Since n does not have units digit 0, the units digits of n and $19n$ will add to 10, so the average in each pair is 5, and thus the overall average in this case is 5. Our final answer is $0 \cdot \frac{1}{4} + 5 \cdot \frac{3}{4} = \frac{15}{4}$.

---

10. Grant thinks of a set of 19 integers. He writes down all products of (not necessarily distinct) pairs of integers from the set. What is the fewest number of distinct values that Grant could have written down?

Solution: The answer is $\boxed{35}$.

By thinking of

$$\{-2^9, \ldots, -2^1\} \cup \{0\} \cup \{2^1, \ldots, 2^9\},$$

Grant writes down $\{-2^{18}, \ldots, -2^2\} \cup \{0\} \cup \{2^2, \ldots, 2^{18}\}$, for a total of 35 numbers. We now show that he always writes at least 35 numbers.

Given a set of i positive integers $a_1 < \cdots < a_i$, and another set of j positive integers $b_1 < \cdots < b_j$, there are at least $i + j - 1$ products of the form $a_k b_\ell$: just consider the $i + j - 1$ products

$$a_1b_1 < a_1b_2 < \cdots < a_1b_j < a_2b_j < a_3b_j < \cdots < a_ib_j.$$

Similarly, given a set of i positive integers and j negative integers, there are at least $i + j - 1$ products between one number from each set.

Now, suppose Grant is thinking of x negative numbers and y negative numbers.

- If 0 is also one of the numbers he is thinking of, $x + y = 18$. He is thinking of at at least $\max(2x - 1, 2y - 1) \geq 17$ positive numbers and $x + y - 1 = 17$ negative numbers, as well as 0. This amounts to at least 35 numbers.
- If 0 is not one of the numbers he is thinking of, $x + y = 19$. Then, $\max(2x - 1, 2y - 1) \geq 19$, so this amounts to at least 37 numbers.

So, Grant needs at least 35 numbers.

---

11. Every second, a real number p is randomly chosen in the interval $[0, 1]$. Then, with probability p, the process continues; otherwise, the process terminates and no more numbers are chosen. What is the probability that, after the process terminates, an odd number of the chosen numbers are greater than $\frac{1}{2}$?

Solution: The answer is $\boxed{\frac{2}{5}}$.

Let x be the answer to the question; then, there is a $1-x$ chance that after the process terminates, an even number of the chosen numbers are greater than $\frac{1}{2}$. If the first probability chosen p_1 is less than $\frac{1}{2}$, there is a $\frac{3}{4}$ chance the process terminates, and a $\frac{1}{4}$ chance that the process continues. On the other hand, if $p_1 > \frac{1}{2}$, there is a $\frac{1}{4}$ chance the process terminates, and a $\frac{3}{4}$ chance that the process continues. So, we have

$$x = \left(\frac{1}{2}\right)\left(\frac{1}{4}\right)x + \left(\frac{1}{2}\right)\left(\frac{1}{4}\right) + \left(\frac{1}{2}\right)\left(\frac{3}{4}\right)(1-x).$$

Solving gives $x = \frac{2}{5}$.

---

12. How many triples (a, b, c) of nonnegative integers are there satisfying

$$\begin{cases} a+b & \leq 12, \\ b+c & \leq 12, \\ c+a & \leq 12? \end{cases}$$

Solution: The answer is $\boxed{616}$.

We split into cases based on how many of the variables are equal to each other.

- Case 1: all three variables are distinct.
  We first consider triples with $a > b > c$. Then, for $a = 2, \ldots, 6$ we have $\binom{a}{2}$ choices for b and c, since b and c can be any nonnegative integers less than a. For $a = 7, \ldots, 11$ we have $\binom{13-a}{2}$ choices for b and c, since b and c can be at most $12 - a$. In this case, there are therefore

  $$2\left(\binom{2}{2} + \cdots + \binom{6}{2}\right) = 70$$

  triples with $a > b > c$, and $70 \cdot 3! = 420$ triples altogether.
- Case 2: two of the three variables are equal. First we look at the case when $a = b \neq c$. For $a = 0, \ldots, 6$, c can be any integer from 0 to $12 - a$, excluding a, which gives $12 - a$ choices for c. In this case, there are therefore

  $$12 + 11 + \cdots + 6 = 63$$

  triples with $a = b \neq c$. The $a = c$ and $b = c$ cases are the same, so there are $63 \cdot 3 = 189$ triples altogether.
- Case 3: all three variables are equal. There are evidently 7 triples.

Our final answer is $420 + 189 + 7 = 616$.

---

13. There are two lines tangent to the parabolas $y = x^2 + 4x + 9$ and $y = 4x^2 - 8x + 3$. At what point do they meet?

Solution: The answer is $\boxed{(2, -3)}$.

Rewrite the equations in the form

$$\begin{cases} y &= (x+2)^2 + 5 \\ 4y &= (4x-4)^2 - 4. \end{cases}$$

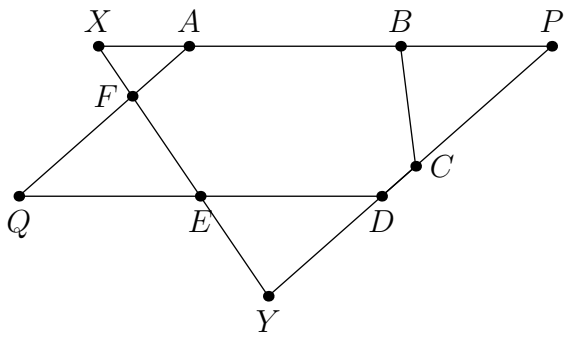
With the second parabola written in this form, it is clear that it is a $\times\frac{1}{4}$ positive dilation of the original parabola. Then, it follows that the tangent lines will meet at this center of dilation.

To find the center of dilation, we consider the vertices of the respective parabolas. The first parabola has vertex $P = (-2, 5)$ while the second parabola has vertex $Q = (1, -1)$. The center of dilation is the point R on ray $\overrightarrow{PQ}$ past Q with $PQ : QR = 3 : 1$, so $R = (2, -3)$, as desired.

---

14. Let $ABCDEF$ be a hexagon with $AB = 14$, $BC = 8$, $CD = 3$, $DE = 12$, $EF = 8$, and $FA = 5$. Suppose $AB \parallel DE$ and $CD \parallel AF$. Line EF meets AB at X and CD at Y. Given that $XY = 20$, what is the area of $ABCDEF$?

Solution: The answer is $\boxed{60\sqrt{7}}$.



Construct parallelogram $AQPD$ as shown in the diagram below. As $AQ + QD = PD + AP$ and $AF + DE = 17 = AB + CD$, we get $FQ + QE = CP + PB$. This, along with $\angle FQE = \angle BPC$ and $FE = BC$, is enough to force $\triangle BPC \cong \triangle FQE$.

Hence let $FQ = BP = x$, so $QE = CP = x + 2$. Next, $\triangle XFA \sim \triangle EFQ \sim \triangle EYD$ gives that $EY = \frac{DE \cdot FE}{QE} = \frac{96}{x+2}$ and $XF = \frac{AF \cdot FE}{FQ} = \frac{40}{x}$. Hence $20 = XF + FE + EF = 8 + \frac{40}{x} + \frac{96}{x+2}$ which solves to yield $x = 10$. Hence triangles FQE and BPC have side lengths $8, 10, 12$ so by Heron's formula they

each have area $\sqrt{15 \cdot 7 \cdot 5 \cdot 3} = 15\sqrt{7}$. Moreover, triangle AQD has area $15\sqrt{7} \cdot \frac{15}{10} \cdot \frac{24}{12} = 45\sqrt{7}$. Thus hexagon $ABCDEF$ has area

$$[APQD] - [QFE] - [BPC] = 90\sqrt{7} - 15\sqrt{7} - 15\sqrt{7} = 60\sqrt{7}.$$

---

15. In an 8 by 8 grid, an *up-right* path is a sequence of 15 cells which begins in the bottom-left cell and ends in the top-right cell, with each cell being directly above or to the right of the previous one. How many sets of 8 distinct up-right paths are there such that exactly one path traverses each of the 36 cells not on the edge of the grid?

Solution: The answer is $\boxed{4096}$.

Consider the 8 cells along the diagonal from the top left to the bottom right cell. By the condition, each of these squares is traversed by exactly one path (the condition doesn't apply to the corners, but there is only one unique path passing through each of those cells). Then we only have to find the number of sets of 8 paths from the bottom left corner to each of the squares on the main diagonal, then we can square that number to account for choosing the paths from the main diagonal to the top right.

Starting at a non-corner square on the main diagonal, we can choose a direction for the previous square along its path, left or down. Then all other non-corner squares must have arrived from the same direction or else the exactly one path condition will fail. The paths from corner squares must be either fully vertical or fully horizontal. Then this reduces to the same situation with a shorter diagonal. The same logic applies 6 times in total, so 6 choices of left or down determine the set of paths uniquely. This gives 64 ways, which we square to get the final answer of 4096.



Chapter 4

Guts Test Solutions



4.1 Round 1

1. While baking croissants, Harini realizes that there is a unique positive integer whose divisors sum to 68. What is this integer?

   Solution: The answer is $\boxed{67}$.

   Since 67 is prime, the divisors of 67 sum to $67 + 1 = 68$. We are given that the answer is unique, so 67 must be the answer.

---

2. Evan distributes 80 indistinguishable candles across 9 distinguishable cakes. If each cake can have at most 9 candles, how many ways can the candles be placed?

   Solution: The answer is $\boxed{9}$.

   Eight cakes must have 9 candles, with one cake having 8 candles. There are 9 choices for which cake receives 8 candles, so the answer is 9.

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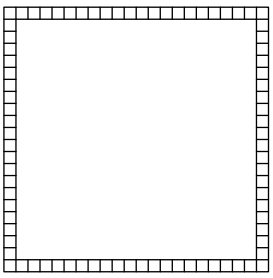
3. Grandma uses 1 part sugar, 2 parts butter and 3 parts flour to bake shortbread cookies. However, before she can make the cookies, Johnny eats 8 grams of sugar. Grandma makes cookies until she runs out of sugar. How many grams of extra ingredients is she left with?

   Solution: The answer is $\boxed{40}$.

   The 8 grams of sugar that Johnny ate would have been used with 16 grams of butter and 24 grams of flour to make the cookies, as per Grandma's $1 : 2 : 3$ recipe. Now that Johnny has eaten the sugar, $16 + 24 = 40$ extra grams of ingredients will remain.

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4. Grant bakes brownies for his 85 friends in a unit square tray. He cuts the brownies into 85 square pieces, as shown below. What is the area of the largest piece?



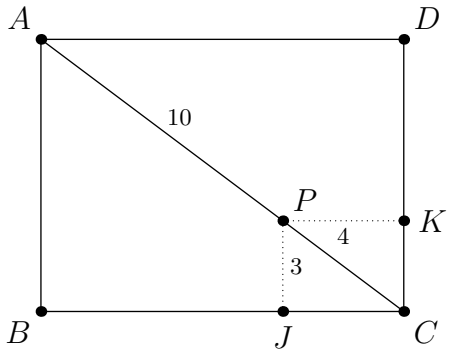
Solution: The answer is $\boxed{\frac{100}{121}}$.

Each of the 84 small square pieces must be of equal size. There are $\frac{84}{4}+1=22$ pieces along each side of the tray, so each small piece has side length $\frac{1}{22}$. Hence, the large piece has side length $20\cdot\frac{1}{22}=\frac{10}{11}$. Squaring this gives us the area, $\frac{100}{121}$.

4.2 Round 2

5. On a rectangular page $ABCD$, Celine marks point P lies on segment $\overline{AC}$ such that $AP=10$. If the distances from P to sides $\overline{BC}$ and $\overline{CD}$ are 3 and 4, respectively, what is the area of $ABCD$?

Solution: The answer is $\boxed{108}$.



Let J and K be the feet from P to sides $\overline{BC}$ and $\overline{CD}$, respectively. By the Pythagorean theorem, $PC=\sqrt{3^2+4^2}=5$. So, a $3\times$ dilation at C will map rectangle $PJCK$ to $ABCD$. Hence, the area of $ABCD$ is

$$[PKCJ]\times 3^2=108.$$

---

6. For homework, Stanley is assigned reading from pages A to B. Stanley does $\frac{1}{6}$ of the homework, reading from pages 202 to 214. What is $B-A$?

Solution: The answer is $\boxed{77}$.

Stanley read $214-202+1=13$ pages, so his homework was $6\times 13=78$ pages. So, $B-A+1=78$, meaning $B-A=77$.

---

7. Evan reads 35 pages of a book, starting somewhere in the middle. Then, Albert reads the next 30 pages. They discover that the sum of the page numbers they read were the same. What was this sum?

Solution: The answer is $\boxed{6825}$.

We can solve this problem by letting x be the first page number that Evan read and solving the resulting equation. However, we present a more creative approach.

If Albert had read 5 more pages, he would have also read 35 pages. Then, the ith page that Albert read would have been 35 pages later than the ith page that Evan read, so the sum of their page numbers would have differed by $35^2 = 1225$. So, the next 5 pages after the ones that Albert read have page numbers summing to 1225, and hence an average of 245. Now, by the same logic as 2024 EMCC Speed #10, the average of the 30 pages that Albert *did* read equals

$$245 - \frac{30+5}{2} = 227.5.$$

Multiplying by 30 gives us our final answer.

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8. A 12-hour digital clock is in sync with an analog clock. While reading, Alice observes that the minute and hour hands of the analog clock point in exactly opposite directions. What hour must it *not* show on the digital clock? (The hands of the analog clock move continuously.)

Solution: The answer is $\boxed{5}$.

In the 12-hour interval from 12 : 00 am to 11 : 59 am, the hour hand makes 1 revolution while the minute hand makes 12. Hence, the minute hand passes the hour hand 11 times every 12 hours. So, the minute hand is directly opposite the hour hand 11 times every 12 hours as well. Moreover, since the minute hand rotates linearly from the frame of reference of the hour hand, these 11 moments are equally spaced. (That is, they occur $\frac{12}{11}^{\text{ths}}$ of an hour apart.)

Now, note that at 6 : 00 am sharp, the minute and hour hand are opposite each other. The last time before this when the hands were opposite was $\frac{12}{11}^{\text{ths}}$ of an hour earlier, which would have been before 5 : 00 am. Hence, at no point from 5 : 00 to 5 : 59 am were the hands opposite each other.

4.3 Round 3

9. Let n be a positive integer. Let G be a divisor of n and let D be a divisor of $n+1$. If $GD = 225$, what is the minimum possible value of n?

Solution: The answer is $\boxed{99}$.

By the Euclidean algorithm, we have $\gcd(n, n+1) = \gcd(n, 1) = 1$. Hence, G and D can share no divisors other than 1. Since

$$GD = 225 = 3^2 \cdot 5^2,$$

it follows that G and D are 9 and 25 in some order. We now consider both cases:

- If $G = 9$ and $D = 25$, for some positive integers j and k, we have $25k - 9j = 1$ and $9j = n$. We may find using trial and error that $(j, k) = (11, 4)$ is a solution, so all solutions are of the form $(11 + 25\ell, 4 + 9\ell)$. Hence, the minimum possible value of n in this case is $9 \times 11 = 99$.
- If $G = 25$ and $D = 9$, for some positive integers j and k, we have $9k - 25j = 1$ and $25j = n$. We may find using trial and error that $(j, k) = (5, 14)$ is a solution, so all solutions are of the form $(5 + 9\ell, 14 + 25\ell)$. Hence, the minimum possible value of n in this case is $5 \times 25 = 125$.

Since $99 < 125$, our final answer is 99.

---

10. Rob announces the schedule for the next four versions of Geometry Dash:

Version	Date
2.3	June 21st, 2026
2.4	November 4th, 2026
2.5	November 19th, 2027
2.6	June 7th, 2028

If X is the number of days between the release of 2.3 and 2.4, and Y is the number of days between the release of 2.5 and 2.6, what is $X + Y$?

Solution: The answer is $\boxed{337}$.

Let X' be the number of days between June 7th, 2027 and November 4th, 2027. Let Y' be the number of days between November 4th, 2027 and June 7th, 2028. We note the following relations:

- We have $X' - X = 14$, as the interval of dates within X' is the same as that of X, but shifted one year forward with the starting date moved 14 days forwards.
- We have $Y' - Y = 15$, as the interval of dates within Y' is the same as that of Y, but with the starting date moved 15 days forwards.
- We have $X' + Y' = 366$, as the combined intervals of X' and Y' span exactly a year, including the leap day of February 29th, 2028.

So,

$$X + Y = (X' + Y') - (X' - X) - (Y' - Y) = 366 - 14 - 15 = 337,$$

which is our final answer.

---

11. Benny is playing Clubstep. In each attempt, he either beats the level or dies at some integer percentage from 1% to 99%. If he starts at 0%, for each integer $1 \leq x \leq 99$, he has an $x\%$ chance of dying $x\%$ or before in the level. If Benny starts playing Clubstep from the start of 97% (the moment the progress bar moves from 96% to 97%), what is the probability he dies at 98%? Express your answer as a fraction.

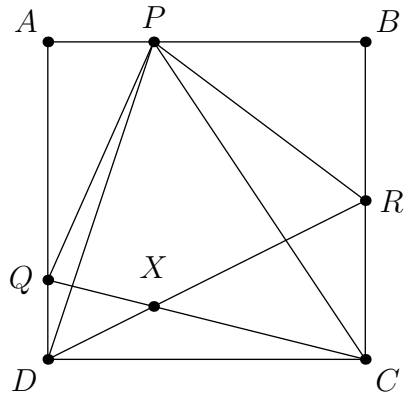
Solution: The answer is $\boxed{25\% \text{ or } \frac{1}{4}}$.

There are 100 possible events when Benny starts from 0%, each of which has a 1% chance of occuring: either Benny dies at $1\%, 2\%, \ldots, 99\%$, or Benny beats the game. When starting from 97%, we know that the first 96 events can't happen, and the remaining four events will occur with equal liklihood. So, the answer is $\frac{1}{4}$.

---

12. Albert is designing a custom cube (square). He begins with square $ABCD$ and selects a point P on side $\overline{AB}$ such that $AP = 2$ and $PB = 4$. He then adds points Q and R on sides $\overline{DA}$ and $\overline{BC}$, respectively, such that the areas of $\triangle DPR$ and $\triangle CPQ$ are both equal to 15. If lines $\overline{CQ}$ and $\overline{DR}$ meet at X, what is PX?

Solution: The answer is $\boxed{5}$.



Let J be the point on line CQ for which $\overline{JP} \perp \overline{AB}$. Let E and F be the feet from Q and C to line PJ, respectively. We have

$$15 = [PQC] = [PJQ] + [PJC] = \frac{PJ}{2}(QE + CF) = \frac{PJ}{2} \cdot 6,$$

so $PJ = 5$. Likewise, if we let K be the point on line DR for which $\overline{KP} \perp \overline{AB}$, we have $PK = 5$. So, $J = K$, and this common point is the intersection of lines DR and CQ, or equivalently, X. So, $PX = 5$, as desired.

4.4 Round 4

13. Angelica creates an infinite string of integers by writing $1, 2, 3, \ldots$ in a row as follows:

$$1234567891011121314\ldots$$

What five digits appear right before the first occurrence of the substring 54345?

Solution: The answer is $\boxed{45334}$.

We claim 54345 is first formed when Angelica writes down

$$\ldots, 3453, 3454, 3455, \ldots$$

The substring 54345 does not contain two consecutive single digital or double digit numbers. Moreover, 54345 can't be taken from a neighborhood of three digit numbers; otherwise, either 543, 434 or 345 would be one of the three digit numbers, but we can check that these numbers' neighborhoods look different.

So, the first occurence of 54345 must be in a four-digit neighborhood. Since these five consecutive digits must contain the leftmost digit of some four-digit number, this digit must be at least 3. In this case, we know the first three digits of the second number are 345, and that the last two digits of the previous number are 54. So, the second number must be 3455, allowing us to recover the answer.

---

14. A debate tournament contains 5 students who are ranked $1, 2, \ldots, 5$. Pat does not know the ranking of the students. She randomly chooses six pairs of players without replacement, and is told the higher ranked student among each pair. What is the probability that this information allows Pat to determine each student's ranking?

Solution: The answer is $\boxed{\frac{1}{14}}$.

We claim Pat can determine each student's ranking if and only if the pairs 12, 23, 34 and 45 happen to be among her queries. Call such a pair "consetocive."

If Pat does end up querying every consetocive pair, she will know because she has a chain of length 5 among her query results; this chain can only correspond to the consetocive pairs, and Pat can determine the rankings.

Otherwise, if Pat does not query some consetocive pair $(i, i+1)$, Pat has no information to distinguish between the ranking of student i and $i+1$.

The claim follows. To count the answer, note that there are $\binom{5}{2} = 10$ pairs total; there are $\binom{10-4}{6-4}$ ways for Pat to choose 6 pairs including the four consetocive ones. So, our final answer is

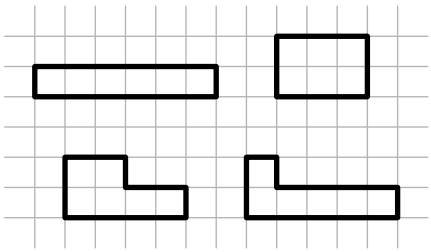
$$\frac{\binom{6}{2}}{\binom{10}{6}} = \frac{1}{14}.$$

---

15. A *hexomino* is a shape made from 6 connected unit squares. Grant's job is to partition a 2×60 rectangle into congruent hexominos. How many ways can he accomplish this? Partitions which differ by a rotation or reflection are considered distinct.

Solution: The answer is $\boxed{2050}$.

There are four noncongruent hexominos that can fit inside Grant's retocangle, pictured below.



Clearly, there is 1 way to tile Grant's retocangle with each of the first two hexominos. In order to tile the retocangle with either of the other two hexominos, Grant must subdivide the retocangle into ten 2×6 retocangles, and pick one of two orientations to fit two hexominos in each of the 2×6 retocangles. So, there are 2^{10} ways for Grant to fit the hexominos in each of these cases. our final answer is

$$1 + 1 + 2^{10} + 2^{10} = 2050.$$

---

16. Lia places her wireless earbuds at fixed points B and C on the plane with $BC = 1$. Let A be a moving point, let M be the midpoint of $\overline{AB}$ and let G be the centroid of $\triangle ABC$. What is the area of the region formed by points A for which $AM \leq GM$?

Solution: The answer is $\boxed{\frac{9}{16}\pi}$.

Fix $B = (0, 0)$ and $C = (1, 0)$. Let $A = (x, y)$. Then, $M = (\frac{x}{2}, \frac{y}{2})$ and $G = (\frac{x+1}{3}, \frac{y}{3})$. The condition $AM \leq GM$ is equivalent to

$$AM^2 \leq GM^2$$
$$\left(x - \frac{x}{2}\right)^2 + \left(y - \frac{y}{2}\right)^2 \leq \left(\frac{x+1}{3} - \frac{x}{2}\right)^2 + \left(\frac{y}{3} - \frac{y}{2}\right)^2$$
$$\left(x + \frac{1}{4}\right)^2 + y^2 \leq \frac{9}{16}.$$

So, the valid points A lie in a circle with radius $\frac{3}{4}$, giving us our final answer.

4.5 Round 5

17. Alice finds the area of a square with side length 17^5 (in base ten) and converts it to a base twelve numeral. What is the sum of the digits of this area in the base-twelve representation? (Express your answer in base ten.)

Solution: The answer is $\boxed{56}$.

We want to find the base 12 representation of 17^{10}. Note that

$$17^{10} = 289^5 = (2x+1)^5,$$

where $x = 12^2$. We can then expand this, with coefficients written in base 12:

$$28x^5 + 68x^4 + 68x^3 + 34x^2 + Ax + 1.$$

Substituting back $x = 12^2$, we see that our final number is

$$286868340A01,$$

whose digits can be checked to sum to 56.

---

18. How many sets of ten consecutive positive integers exist such that their product is a multiple of the square of the largest number?

Solution: The answer is $\boxed{151}$.

This condition is equivalent to the tenth number dividing the product of the other 9. Letting the tenth number be n, we have

$$(n-1)(n-2)\cdots(n-9) \equiv 0 \pmod{n} \qquad \Longrightarrow \qquad -9! \equiv 0 \pmod{n},$$

so $n \mid 9!$. We can check that $9!$ has 160 divisors. Moreover, since $n \geq 10$ and each of 1 through 9 divide $9!$, there are $160 - 9 = 151$ possible values of n.

---

19. How many ways can Alice fill in each $\square$ with a $+$, $-$, $\times$ or $\div$ such that the following equation is true? (She cannot add parentheses.)

$$1 \square 1 \square 1 \square 1 \square 1 \square 1 \square 1 = 2.$$

Solution: The answer is $\boxed{792}$.

Between addition and subtraction signs, all expressions will equal 1 (since multiplying or dividing by 1 does nothing). So, it suffices to count the number of ways to write i addition or subtraction signs for each i, and multiply this by 2^{6-i}.

We must either have one + and no −, two + and one − or three + and two −. There are $\binom{6}{1,0,5}$, $\binom{6}{2,1,3}$ and $\binom{6}{3,2,1}$ ways to arrange the position of these + and − signs, respectively. Multiplying by their respective powers of 2, we get our final answer of

$$\binom{6}{1,0,5} \cdot 2^5 + \binom{6}{2,1,3} \cdot 2^3 + \binom{6}{3,2,1} \cdot 2^1 = 792.$$

---

20. Let $ABCD$ be a unit square. Points E and F lie on segments $\overline{CD}$ and $\overline{BD}$, respectively, such that $CF = CE$ and $\angle EAF = 45°$. What is AE?

Solution: The answer is $\boxed{\sqrt{6} - \sqrt{2}}$.

We claim that $\triangle CEF$ is equilateral. Indeed, since $\angle AFE = \angle ADE = 90°$, quadrilateral $ADEF$ is cyclic. Moreover, since $\overline{DF}$ bisects $\angle ADE$, it follows that $AF = FE$. By symmetry in square $ABCD$, we also have $AF = CF$, so $CF = FE$. Combined with $CF = CE$ given in the problem, the claim follows.

Let $AF = x$ and let X be the foot from F to line CD. Since $\triangle FXC$ and $\triangle FXD$ are 30-60-90 and 45-45-90 respectively, we have

$$1 = CX + DX = \frac{x}{2} + \frac{x\sqrt{3}}{2},$$

so $x = \sqrt{3} - 1$. Then, $AE = x\sqrt{2} = \sqrt{6} - \sqrt{2}$, as desired.

4.6 Round 6

21. What ordered pair (x, y) of positive real numbers satisfies

$$\frac{y^2}{x^2} = \frac{x-1}{y-2} = \frac{2x+2}{2y+5}?$$

Solution: The answer is $\boxed{(\frac{3}{19}, \frac{2}{19})}$.

Since $\frac{2x-2}{2y-4} = \frac{2x+2}{2y+5}$, both expressions must be equal to $\frac{4}{9}$; therefore, $\frac{x}{y} = \frac{3}{2}$. Let $x = 3t$ and $y = 2t$. We have

$$\frac{3t-1}{2t-2} = \frac{4}{9} \qquad \Longrightarrow \qquad t = \frac{1}{19},$$

allowing us to extract the answer.

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22. Let $s_2(x)$ denote the sum of the digits of x when expressed in binary. How many pairs of integers (m, n) exist such that $0 \leq m < n \leq 127$ and $s_2(m) > s_2(n)$?

Solution: The answer is $\boxed{1366}$.

Each of m and n can be represented as a 7-bit binary string. The key idea is that the bits at which m and n agree can be ignored. Suppose the binary strings agree at $i \leq 6$ of the 7 bits. Then

- there are $\binom{7}{i}$ ways to choose which of the bits they agree at;
- there are 2^i ways to choose whether each of these agreements is a 0 or a 1;
- there is 1 way to choose the leftmost disagreement. Indeed, since $m < n$, in the first disagreeing bit, m must have a 0 and n must have a 1.
- There are
$$\binom{6-i}{\lceil (6-i)/2 \rceil + 1} + \cdots + \binom{6-i}{6-i}$$
ways to pick the remaining $6-i$ disagreements, since strictly more than $\lceil \frac{6-i}{2} \rceil$ of the disagreements must go to m by the bit sum condition.

So, our final answer is

$$\sum_{i=0}^{6} \left(\binom{7}{i} \right) (2^i) \left(\binom{6-i}{\lceil (6-i)/2 \rceil + 1} + \cdots + \binom{6-i}{6-i} \right) = 1366.$$

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23. Let $PABCD$ be a pyramid such that base $ABCD$ is a rectangle. Suppose the areas of faces PAB, PBC, PCD, PDA and $ABCD$ are 10, 13, 11, 14 and 6, respectively. What is AB/BC?

Solution: The answer is $\boxed{\frac{\sqrt{15}}{5}}$.

Let W, X, Y and Z be the feet from P to lines AB, BC, CD and DA, respectively.

We have that $[PAB] : [PCD] : [ABCD] = 10 : 11 : 6$; dividing the left hand side by AB, it follows that

$$WP : PY : YW = 10 : 11 : 3.$$

Now, by Heron's formula, the distance from P to line WY is $(\frac{WY}{3})(4\sqrt{6})$.

Similarly, $XP : PZ : ZX = 13 : 14 : 3$, and by Heron's formula, the distance from P to line ZX is $(\frac{XZ}{3})(4\sqrt{10})$.

But the distances from P to lines WY and ZX are equal (both are equal to the distance from P to plane $ABCD$). So,

$$\left(\frac{WY}{3}\right)\left(4\sqrt{6}\right) = \left(\frac{XZ}{3}\right)\left(4\sqrt{10}\right) \qquad \Longrightarrow \qquad \frac{AB}{BC} = \frac{XZ}{WY} = \frac{\sqrt{15}}{5}.$$

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24. Let a and b be positive integers. If there is a unique pair of nonnegative integers (x, y) such that $ax + by = 2026$, what is the minimum possible value of $a + b$?

Solution: The answer is $\boxed{67}$.



4.7 Estimation Round

25. Estimate the number of sets of positive integers with an average of 10.
Submit a positive integer E. If the correct answer is A, you will receive $\lfloor 5.9 \min(\frac{E}{A}, \frac{A}{E}) \rfloor$ points.

Solution: The answer is $\boxed{196239}$.

26. Let $s_{10}(x)$ denote the sum of the digits of x when expressed in base 10. Estimate the number of pairs of integers (m, n) with $0 \leq m \leq n \leq 127$ and $s_{10}(m) > s_{10}(n)$.
Submit a positive integer E. If the correct answer is A, you will receive $\lfloor 5.9 \min(\frac{E}{A}, \frac{A}{E})^3 \rfloor$ points.

Solution: The answer is $\boxed{2861}$.

27. Estimate the total number of numerical digits that appear on the Speed, Accuracy and Team round papers. This is counted by using the `CTRL+F` shortcut on the PDF of each round, and therefore includes digits from items such as page numbers and problem numbers.
Submit a positive integer E. If the correct answer is A, you will receive $\lfloor 5.9 \min(\frac{E}{A}, \frac{A}{E})^2 \rfloor$ points.

Solution: The answer is $\boxed{503}$.

28. Estimate the sum of $\frac{1}{\sqrt{x}}$ over all problem numbers x that ChatGPT 5.2 was unable to solve in one attempt, across all four rounds of EMCC 2026 (excluding the Estimation Round). For example, if you think Chat missed Accuracy #4, Guts #4 and Guts #16, guess 1.25. Note that ChatGPT is allowed to write and run computer programs to solve problems with brute force.
Submit a positive real E. If the correct answer is A, you will receive $\lfloor 5.9 \min(\frac{E}{A}, \frac{A}{E})^3 \rfloor$ points.

Solution: The answer is $\boxed{4.0608}$.

Here are the problems that ChatGPT 5.2 missed:

- **Speed P8**, with an unsimplified (!) answer of $\frac{29}{899}$.
- **Speed P19**, with an incorrect answer of 22.
- **Accuracy P7**, unable to arrive at an answer.
- **Team P1**, with an incorrect answer of 132.
- **Team P4**, with an incorrect answer of 16.
- **Team P10**, with an incorrect answer of 36.
- **Team P14**, unable to arrive at an answer.
- **Team P15**, unable to arrive at an answer.
- **Guts P4**, with an incorrect answer of $\frac{1}{2}$.
- **Guts P15**, with an incorrect answer of 2049.

Of course, ChatGPT does not give the same response every time; on another day, it would probably perform very differently than it did here.

